

Towards using Eye Gaze Redirection in Immersive Reading Tasks for Visual Fatigue Reduction

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ABSTRACT

Visual fatigue poses a significant challenge in Human-Information Interaction (HII) in VR. Target redirection, a method that helps mitigate shoulder fatigue, could potentially help reduce visual fatigue by guiding our eyes to move in certain ways. This paper proposes a study using eye movement redirection in VR to reduce visual fatigue from immersive HII tasks. We highlight two primary challenges, speed and range of redirection, that directly impact the user experience and effectiveness. We further introduce two studies aimed at tackling these problems: 1) to find the best parameters for redirection and 2) to evaluate the technique's usability. We hypothesize that this innovative approach will greatly contribute to user-friendly and safe gaze-based HII in VR, enhancing user experience and performance by reducing visual fatigue.

CCS CONCEPTS

• Human-centered computing → Virtual reality.

KEYWORDS

Human-Information-Interaction; Visual Fatigue; Virtual Reality; Cognitive Activity

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1 INTRODUCTION

Virtual Reality (VR) technology has shown significant potential to provide users with immersive experiences through the use of Head-Mounted Displays (HMDs). This evolution introduces a new interaction paradigm in Human-Information-Interaction (HII). In VR, information is presented as virtual objects, and people can access them without constraints of time and location. While disciplines such as Immersive Analytics [18] have proven the utility of VR for professional HII, Apple's recent introduction of the Vision Pro HMD and the spatial computing concept promise to push VR for HII from controlled labs to our offices and living rooms. The increasing integration of VR into everyday HII activities necessitates properly evaluating VR's impact on participants to ensure a safe and effective user experience considering both cognitive and physical aspects [17, 29].

Cognitive fatigue arises from extended periods of concentration, mental overload, and stress, manifesting as reduced concentration and mental exhaustion. In prolonged HII tasks in VR, one of the key characteristics of cognitive fatigue: visual fatigue, will be induced by factors like Vergence-Accommodation Conflict (VAC) [9]. A commonly recognized issue occurs when the human brain receives conflicting signals from the perceived distance of the virtual target and the actual distance to the headset screen [10]. These two types of fatigue share resources and have a symbiotic relationship; visual fatigue can exacerbate cognitive fatigue [1, 11] and vice versa [7], with overlapping symptoms including dry eyes, headaches, and a decrease in task performance [27], further compounding the effects. Finding a solution to reduce visual fatigue and further improve cognitive activities in VR is essential to enhance productivity, comfort, and well-being in HII.

Eye exercises have been proven to be the most effective way to reduce visual fatigue in clinical practice [5, 21]. Common eye exercises include eye movements in various directions and paths (e.g., left-right and up-down turning, figure-eight, and circular rotation) and the change of focus (e.g., thumb movement, pencil push-ups, and the 20-20-20 rule [24]).

While solutions to visual fatigue in real life are well established, techniques to reduce visual fatigue in VR are rarely studied. Previous efforts have mostly been made on the hardware side [15, 16]. For example, adjusting lenses to change the focus [15] and using light-field technology to bring virtual targets to a similar depth as objects in the real world [16]. [30] is the first study to reuse the eye exercise "Figure-eight" in VR interaction and achieved positive results in subjective preference. However, the previous study did not investigate different figure sizes, which can potentially affect the effectiveness of visual fatigue improvement, and implemented the eye exercise during active breaks. Our work addresses this unstudied area, proposing a more natural approach that indirectly nudges participants' eyes to travel along an eight-shape trajectory through a redirected target (See Figure 1).

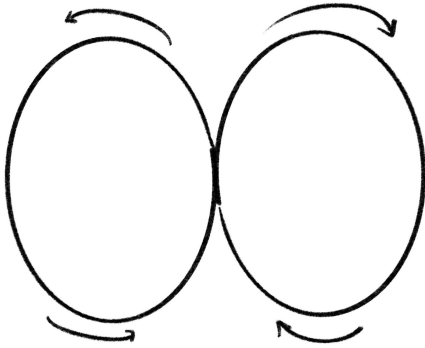


Figure 1: An illustration of figure-eight target redirection trajectory.

VR technologies offer an adjustable workspace free from physical environment constraints. Within this virtual space, participants can position virtual targets for an adaptable and optimized user experience. This flexibility allows for dynamic target placement redirection, indirectly enabling participants to perform eye exercises without interrupting the primary interaction tasks. Target redirection has been widely studied and has been demonstrated to be effective in reducing physical shoulder fatigue in mid-air interaction, as shown in [2, 8, 19]. We hypothesize that eye movement redirection will improve visual fatigue in VR HII tasks and further enhance overall user comfort and productivity.

In this position paper, we present our approach to using eye movement redirection as a novel technique to reduce visual fatigue in VR interactions, contributing new, valuable insights to future engaging, safe, and comfortable HII design.

2 CHALLENGES OF EYE MOVEMENT REDIRECTION

In the current section, we will discuss key considerations for implementing eye movement redirection in the common HII task of *text reading in VR*, focusing on user experience and its effectiveness in reducing cognitive fatigue.

2.1 Head-Bound Coordinate System of Redirected Target

Currently, most VR workspaces adopt a "world-bound" coordinate system (See Figure 2) to place targets with a fixed orientation as it follows the rule of physics in the real world [26]. However, implementing the redirected target in such a system may cause body compensation since participants can rotate their heads or bodies to locate the target. To guarantee the redirected target only affects eye movement, we propose a head-bound coordinate to avoid compensation from other body parts. In this system, the centre of the redirected target is anchored to the centre viewpoint of HMD. As such, no matter what direction participants are facing, the redirected target remains relatively static to participants' head movements (See Figure 3).

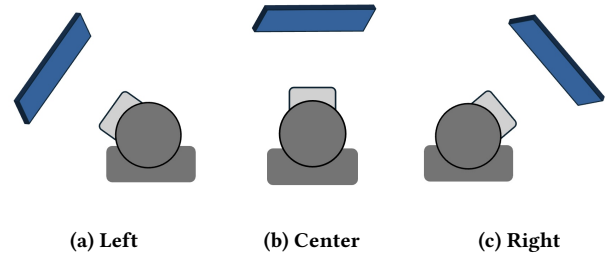


Figure 2: An illustration of world-bound coordinate system.

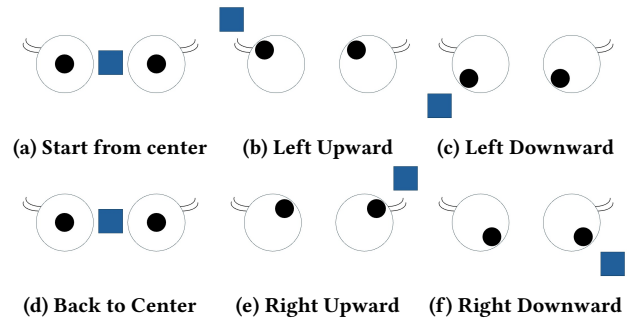


Figure 3: An illustration of redirected eye movement following an eight-shaped trajectory.

2.2 Moving speed of the redirected display

The first consideration when implementing redirected eye movement in immersive reading tasks is the moving speed of the redirected display text. When users visually perceive motion from the interactive target in VR, the discrepancy between their eyes and their stationary bodies might lead to motion sickness, also called cybersickness [3]. Side effects of cybersickness include but are not limited to cognitive fatigue, headaches, eye strain, nausea, and dizziness [23]. Our concern is that implementing redirection will induce additional motion on the displays and further increase the severity

of cybersickness. Therefore, determining the optimal moving speed of the display is critical for minimizing cybersickness associated with the redirected information display in VR and for balancing effective eye movement redirection and user comfort.

On the other hand, the moving speed of the redirected target will affect the level of obtrusiveness of the manipulation. Unobtrusive manipulation has been extensively explored in HCI. The most common applications are redirected touch [14] and redirected walking [22]. The primary motivation for using unobtrusive manipulation is that it maximizes the utilization of physical limitations, such as space [28]. By applying unobtrusive redirection techniques, the virtual space can be deformed relative to the real space. Users can perform their tasks without perceiving the deformation or losing their sense of immersion when all perceptual cues remain consistent. While there are no studies showing a connection between the obtrusiveness level of manipulation and visual fatigue, a high obtrusive level may fasten the occurrence of cybersickness [20]. We will discuss the design of a proposed investigation of the relationship between the moving speed, perception level, and cybersickness of eye movement redirection in Section 3.1.

2.3 Moving range of redirected trajectory

The second key challenge of redirecting eye movement in VR reading tasks is the range of the redirected movement, or in other words, the size of the eight-figure. Thai et al. [30] implemented figure-eight exercises in active breaks during VR engagement and observed improved subjective ratings in alleviating visual fatigue [30]. However, these results were not reflected in the objective visual fatigue measures. In the standard definition of the figure-eight eye exercise [6], it is suggested that the eye changes direction at the end of one full round. Therefore, we hypothesize that including eye movement in the opposite direction of the original movement will increase the effectiveness of reducing visual fatigue. More importantly, the size of the figure was fixed in [30], and the length of the moving path was not investigated, i.e., the distance the eye gaze had to travel. If the figure-eight exercise is not executed properly, for instance, if participants do not follow the movement accurately or the size of the figure-eight does not facilitate sufficient eye movement, the efficacy of the exercise may be compromised [30]. These factors can influence the overall effectiveness of the eye exercise in a VR setting, highlighting the need for careful calibration of the exercise parameters to ensure optimal results.

3 PROPOSED DATA COLLECTION

As explained in Section 2, we need to consider two main challenges when implementing redirection in eye movement: the moving speed and the moving range. Therefore, we propose two studies to investigate these concerns separately. Through these proposed studies, we aim to find a balance between effective eye exercises through eye movement redirection and minimizing the adverse effects of cybersickness.

3.1 Study 1: Finding the optimal parameter of eye movement redirection

We first aim to determine the optimal moving speed for a focus object performing a figure-eight movement. Previous research exploring the impact of different target moving speeds on cybersickness in VR [20] found that the lowest speed level in the proposed study ($1.2m/s$) resulted in minimal cybersickness. However, this study [20] did not investigate speed levels below $1.2m/s$. Based on this, we will first experiment with different levels of moving speed below $1.2m/s$ to observe changes in cybersickness. The four chosen speed levels of dynamic targets are 0.3, 0.6, 0.9, and $1.2m/s$, with a $0.3m/s$ distance in between to ensure a noticeable difference between levels.

Meanwhile, previous work that investigated the perception threshold of object manipulation [25] found that the displacement of a virtual target at a 2 m distance is hard to detect when moving between $0.006 - 0.009m/s$ in both directions. Surprisingly, participants reported significant cybersickness in the study, which suggests that cybersickness can only be reduced, not eliminated. Therefore, we will include one control condition of static targets in the study and collect the pre-intervention cybersickness ratings for baseline comparison. Cybersickness ratings in the current study will be gathered through the Simulator Sickness Questionnaire (SSQ) [12]. The SSQ reports participants' sickness levels across 16 dimensions, such as headache and nausea.

3.2 Study 2: Investigating the effect of redirection in reducing visual fatigue during reading task

In Study 2, we explore the effects of eye movement redirection by applying the optimal moving speed from Study 1. To assess the effectiveness of eye movement redirection to ease visual fatigue, we will test different eight-shaped moving paths and compare them to a baseline condition where the target remains static with no redirection. The task content will be displayed in the screen space to minimise the compensation from neck movements. We propose the choice of the eyeball travelling path based on the field of view (FOV) in the Meta Quest Pro¹. The Meta Quest Pro renders a FOV of 108° horizontally, 95.57° vertically, and 111.24° diagonally, which means the content displayed can be maximized to such a level when the HMD is worn correctly. Thus, we will represent the chosen sizes of the eight-figure in the unit of the maximum FOV (noted as "MAX"), namely 25% MAX, 50% MAX, 75% MAX, and 100% MAX. The task implemented in the study is a text-reading task that is representative of HII. The language of the text will be the local language of the investigating lab, and the content will be chosen from reading materials for elementary school children to minimize the task difficulty and cognitive load to adult participants (See Figure 4).

We will collect both subjective and objective visual fatigue measures to compare the influence of eye movement redirection. The

¹<https://www.meta.com/quest/quest-pro/>



Figure 4: A demonstration of the reading task implemented in the proposed experiments.

subjective measure will be the Computer Vision Syndrome Questionnaire (CVS-Q). CVS-Q provides information about a participant’s usual state of visual fatigue. The authors of CVS-Q[4] suggest that a score above six indicates that a person is likely suffering from visual fatigue. The objective visual fatigue indicator chosen for the study is eye blink rate [13, 16]. We will record participants’ regular blink rates as a baseline before introducing them to the study environment and normalize the continuous blink signal with their baseline measures for comparison between participants. By combining subjective and physiological responses, we expect to gain a comprehensive understanding of how visual fatigue varies between study conditions.

From the proposed Studies 1 and 2, we aim to determine the most effective combination of size and speed of eye movement redirection for reducing visual fatigue in immersive HII tasks. This outcome will help us establish guidelines for eye exercises that enhance comfort and efficiency in VR.

4 EXAMPLE APPLICATIONS OF EYE MOVEMENT REDIRECTION IN VR

Eye movement redirection in VR can significantly enhance the user experience and usability of various applications by reducing visual and cognitive fatigue. In virtual training and education, students and trainees will be able to maintain attention and comprehension for longer periods, improving learning outcomes. Medical professionals using VR for surgical simulations and emergency response training benefit from reduced visual strain, maintaining cognitive sharpness and reducing errors. Remote collaboration and virtual meetings become more effective as users experience less fatigue, leading to better communication and productivity. Designers and

artists working in VR can sustain their creativity and precision for extended periods, while patients undergoing virtual therapy or rehabilitation sessions can stay engaged and benefit more from their exercises. Gamers enjoy longer, more immersive sessions without discomfort, and professionals analyzing complex data can interpret information more accurately. Additionally, the elderly using VR for cognitive stimulation and social interaction can engage more comfortably and beneficially, improving their overall well-being. These improvements collectively enhance the usability and effectiveness of VR, making it a more user-friendly and productive tool across various domains.

5 CONCLUSION AND FUTURE WORK

In this paper, we present our vision for eye movement redirection to improve cognitive fatigue in 3D HII tasks. We describe two main challenges: redirection speed and redirection range, which concern user experience and the effectiveness of redirection. We propose two experiments to determine the optimal parameters of redirection and assess the usability of redirection accordingly. We believe this novel technique will contribute to future gaze-based HII in VR.

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